Instituto Tecnológico de Veracruz



Administración de Base de Datos

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Carrera: Ingeniería en Sistemas Computacionales

Bitacora de CRUD en Neo4j

1. Primero hice instalación de Neo4j en mi computadora
2. Abrí Neo4j Desktop y creé una instancia local llamada Rafael con su contraseña.
3. Inicié la instancia estado RUNNING y me conecté con el usuario neo4j desde Query.
4. Create: creé los nodos de tipo Alumno y Materia y la relación CURSA entre un alumno y la materia:

CREATE (a:Alumno {nombre: "Rafael", edad: 21, carrera: "Sistemas"})

CREATE (b:Alumno {nombre: "Luis", edad: 21, carrera: "Sistemas"})

CREATE (m:Materia {nombre: "Bases de datos NoSQL"})

MATCH (a:Alumno {nombre: "Rafael"}), (m:Materia {nombre: "Bases de datos NoSQL"})

CREATE (a)-[:CURSA]->(m)

5. Read: consulté los alumnos guardados en la base de datos:

MATCH (a:Alumno) RETURN a

6. Update: cambié la edad del alumno Rafael a 21 años:

MATCH (a:Alumno {nombre: "Rafael"}) SET a.edad = 22 RETURN a

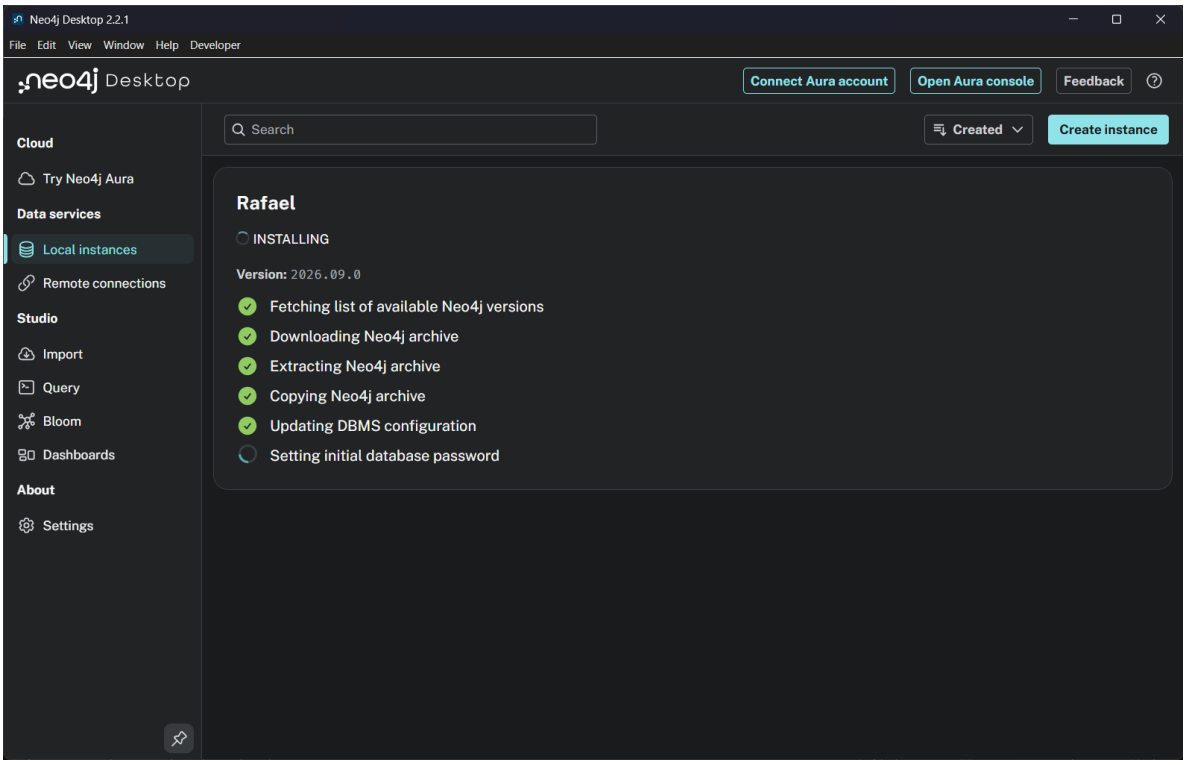
7. Delete: eliminé el nodo de Luis junto con sus relaciones:

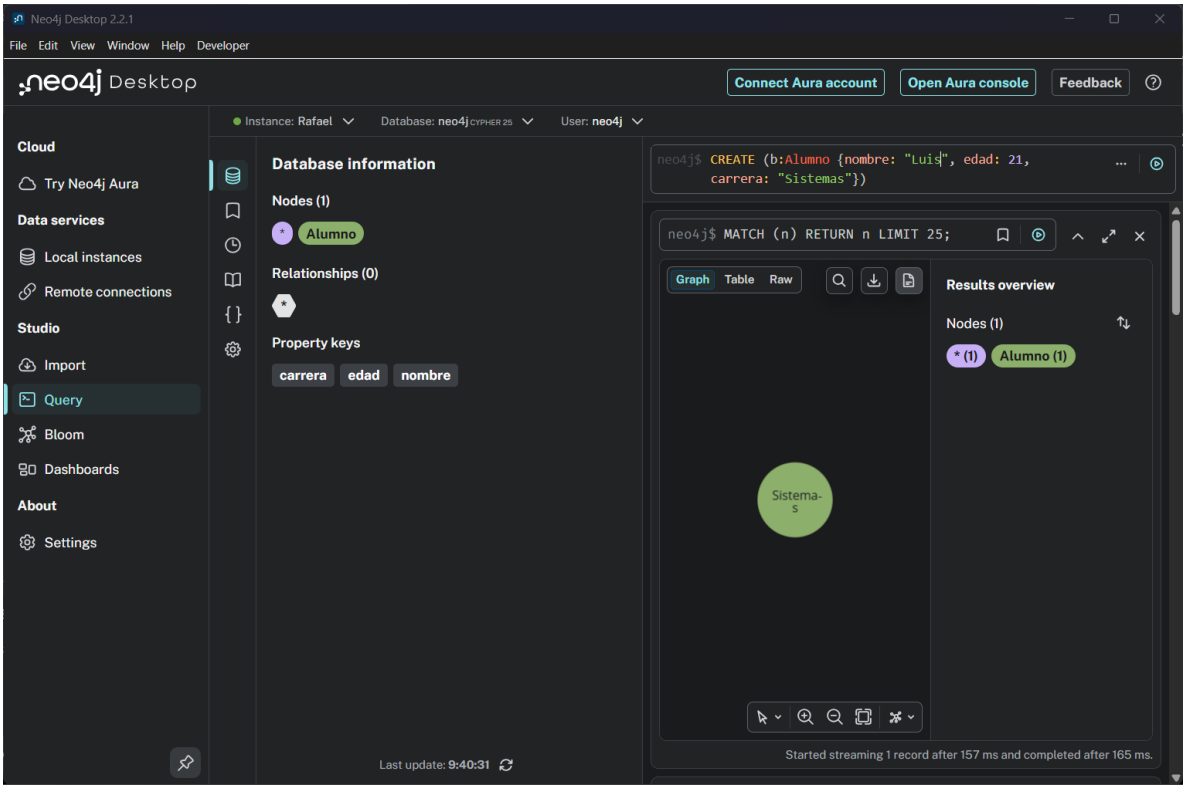
MATCH (a:Alumno {nombre: "Luis"}) DETACH DELETE a

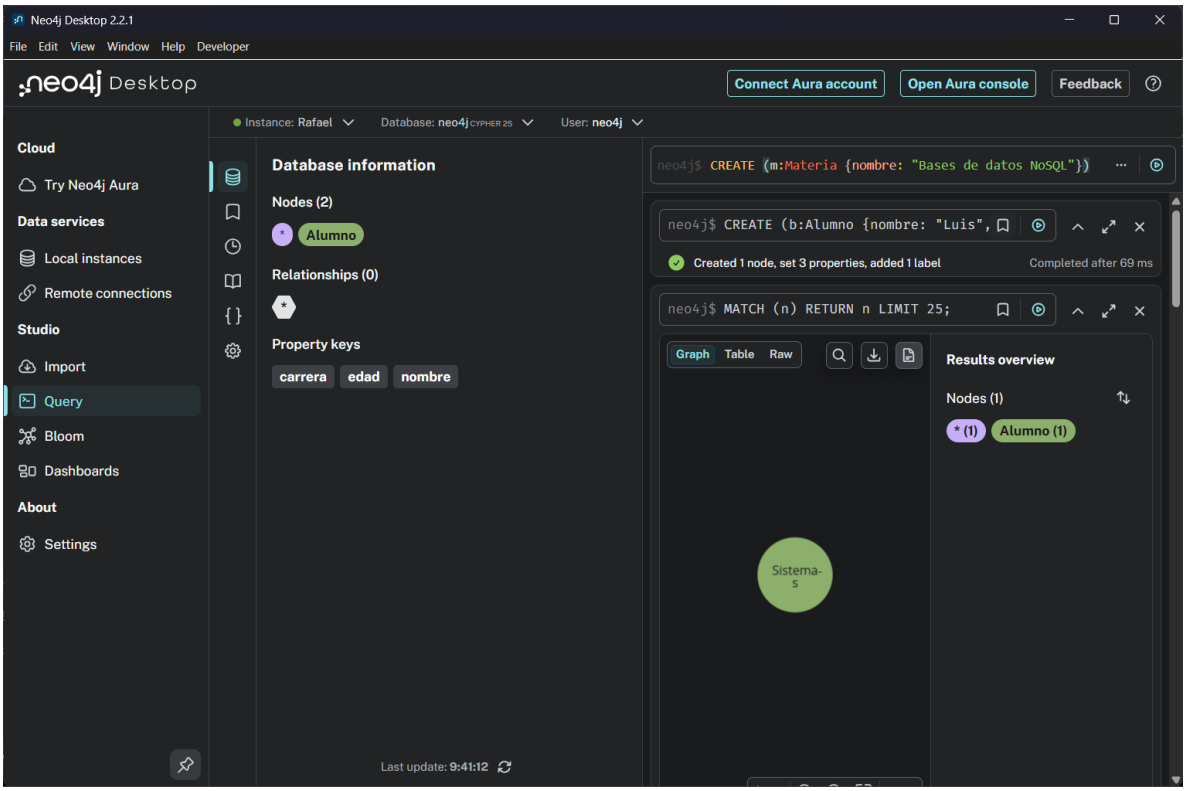
8. Verifiqué el resultado final con MATCH (a:Alumno) RETURN a.nombre, a.edad, donde solo quedó el nodo de Rafael

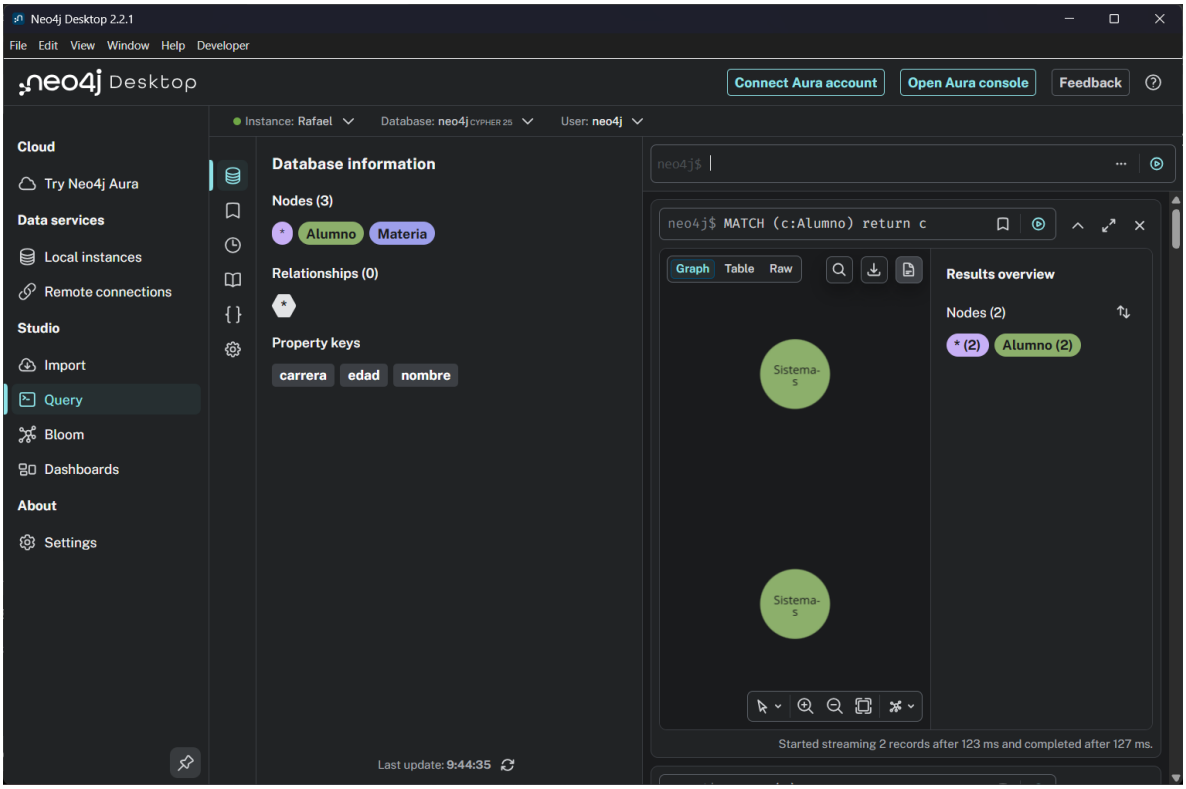
Neo4j realicé correctamente las cuatro operaciones del CRUD: la base de datos terminó con el nodo de Keyla, el nodo de la materia y la relación CURSA entre ambos.

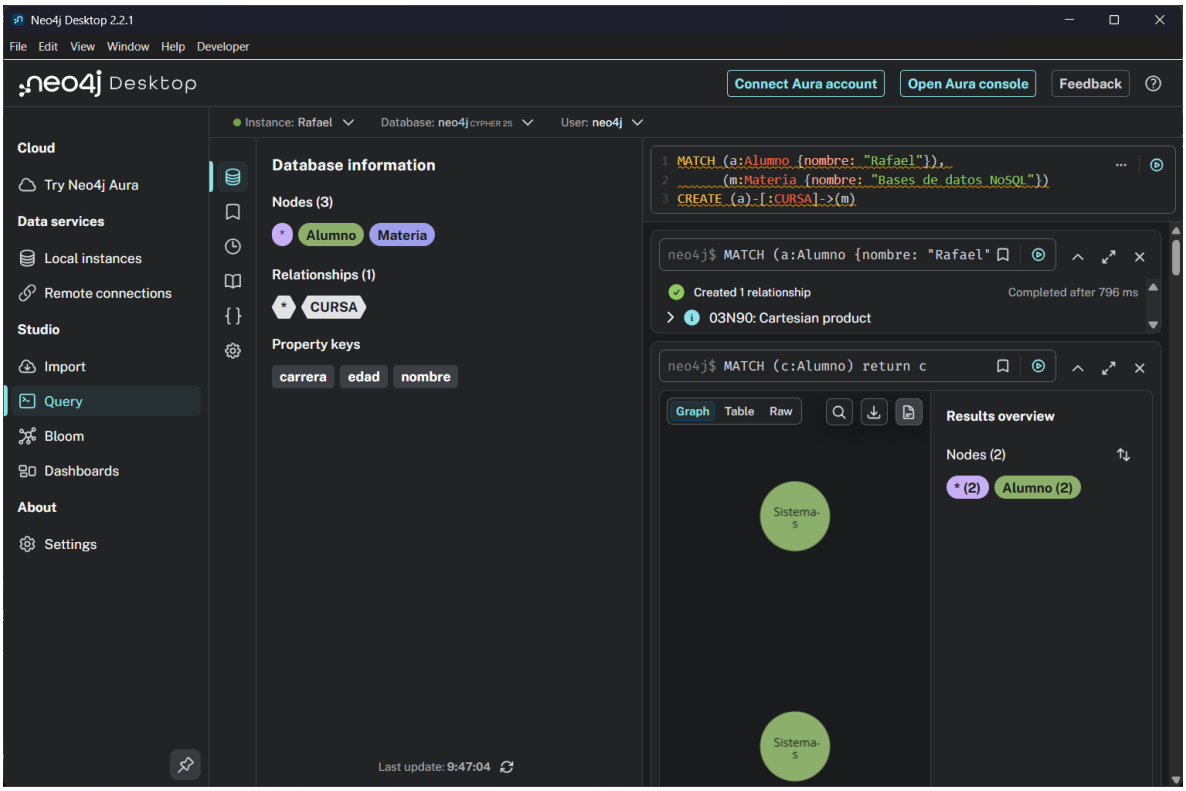
EVIDENCIAS:

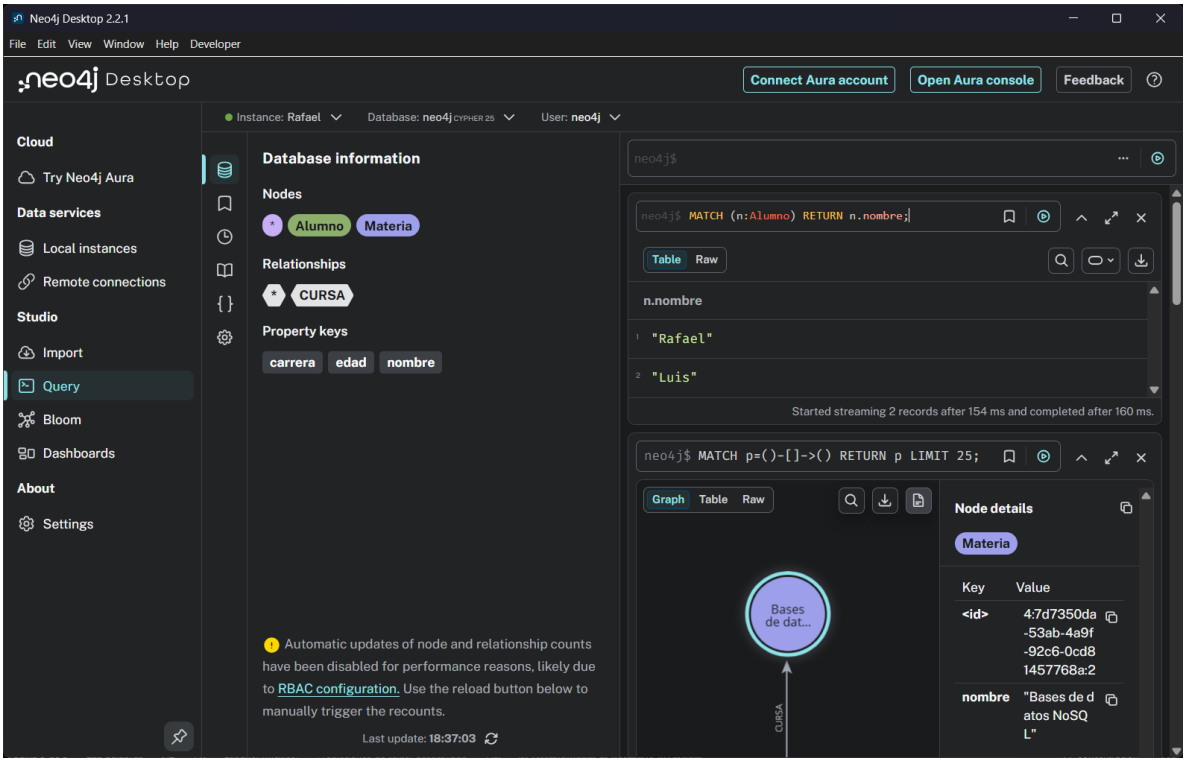

Neo4j Desktop 2.2.1
File Edit View Window Help Developer
neo4j Desktop
Connect Aura account
Open Aura console
Feedback
Cloud
Try Neo4j Aura
Data services
Local instances
Remote connections
Studio
Import
Query
Bloom
Dashboards
About
Settings
Search
Created
Create instance
Rafael
INSTALLING
Version: 2026.09.0
Fetching list of available Neo4j versions
Downloading Neo4j archive
Extracting Neo4j archive
Copying Neo4j archive
Updating DBMS configuration
Setting initial database password


Neo4j Desktop 2.2.1
File Edit View Window Help Developer
neo4j Desktop
Connect Aura account
Open Aura console
Feedback
Instance: Rafael
Database: neo4j CYPHER 25
User: neo4j
Cloud
Try Neo4j Aura
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About
Settings
Database information
Nodes (1)
Alumno
Relationships (0)
Property keys
carrera
edad
nombre
Last update: 9:40:31
neo4j$ CREATE (b:Alumno {nombre: "Luis", edad: 21,
carrera: "Sistemas"})
neo4j$ MATCH (n) RETURN n LIMIT 25;
Graph
Table
Raw
Results overview
Nodes (1)
* (1)
Alumno (1)
Sistema-
s
Started streaming 1 record after 157 ms and completed after 165 ms.


Neo4j Desktop 2.2.1
File Edit View Window Help Developer
neo4j Desktop
Connect Aura account
Open Aura console
Feedback
Instance: Rafael
Database: neo4j CYPHER 25
User: neo4j
Cloud
Try Neo4j Aura
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Remote connections
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Query
Bloom
Dashboards
About
Settings
Database information
Nodes (2)
Alumno
Relationships (0)
Property keys
carrera
edad
nombre
Last update: 9:41:12
neo4j$ CREATE (m:Materia {nombre: "Bases de datos NoSQL"})
neo4j$ CREATE (b:Alumno {nombre: "Luis",
Created 1 node, set 3 properties, added 1 label
Completed after 69 ms
neo4j$ MATCH (n) RETURN n LIMIT 25;
Graph
Table
Raw
Results overview
Nodes (1)
* (1)
Alumno (1)
Sistema-s


Neo4j Desktop 2.2.1
File Edit View Window Help Developer
neo4j Desktop
Connect Aura account
Open Aura console
Feedback
Instance: Rafael
Database: neo4j CYPHER 25
User: neo4j
Cloud
Try Neo4j Aura
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Query
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Dashboards
About
Settings
Database information
Nodes (3)
Alumno
Materia
Relationships (0)
Property keys
carrera
edad
nombre
Last update: 9:44:35
neo4j$
neo4j$ MATCH (c:Alumno) return c
Graph
Table
Raw
Results overview
Nodes (2)
* (2)
Alumno (2)
Sistema-s
Sistema-s
Started streaming 2 records after 123 ms and completed after 127 ms.


Neo4j Desktop 2.2.1
File Edit View Window Help Developer
neo4j Desktop
Connect Aura account
Open Aura console
Feedback
Instance: Rafael
Database: neo4j CYPHER 25
User: neo4j
Cloud
Try Neo4j Aura
Data services
Local instances
Remote connections
Studio
Import
Query
Bloom
Dashboards
About
Settings
Database information
Nodes (3)
Alumno
Materia
Relationships (1)
CURSA
Property keys
carrera
edad
nombre
Last update: 9:47:04
1 MATCH (a:Alumno {nombre: "Rafael"}),
2 (m:Materia {nombre: "Bases de datos NoSQL"})
3 CREATE (a)-[:CURSA]->(m)
neo4j$ MATCH (a:Alumno {nombre: "Rafael"
Created 1 relationship
Completed after 796 ms
03N90: Cartesian product
neo4j$ MATCH (c:Alumno) return c
Graph
Table
Raw
Results overview
Nodes (2)
* (2)
Alumno (2)
Sistema-s
Sistema-s


Neo4j Desktop 2.2.1
File Edit View Window Help Developer
neo4j Desktop
Connect Aura account
Open Aura console
Feedback
Instance: Rafael
Database: neo4j CYPHER 25
User: neo4j
Cloud
Try Neo4j Aura
Data services
Local instances
Remote connections
Studio
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Query
Bloom
Dashboards
About
Settings
Database information
Nodes
Alumno
Materia
Relationships
CURSA
Property keys
carrera
edad
nombre
Automatic updates of node and relationship counts have been disabled for performance reasons, likely due to RBAC configuration. Use the reload button below to manually trigger the recounts.
Last update: 18:37:03
neo4j$ MATCH (n:Alumno) RETURN n.nombre;
Table
Raw
n.nombre
1 "Rafael"
2 "Luis"
Started streaming 2 records after 154 ms and completed after 160 ms.
neo4j$ MATCH p=()-[]->() RETURN p LIMIT 25;
Graph
Table
Raw
Node details
Materia
Key
Value
<id>
4:7d7350da-53ab-4a9f-92c6-0cd81457768a:2
nombre
"Bases de datos NoSQL"
Bases de dat...
CURSA